Toward a four-dimensional Yang–Mills construction

Notebook 22: transport into the next coarse background

Research working document

17 October 2026

Abstract

We construct a strictly supported expansion of the actual physical stationary current background as a function of the next-coarse field, uniformly in preceding classical refinement and periodic volume. An affine version of the exact ordinary pivot chart and the retained flat inverse yield a support-valued stationary equation. Coordinatewise Cauchy estimates convert the previously localized scalar action into an analytic gradient majorant without a factorial loss. The solution is identified with the established gauge-zero classical branch. We then transport holomorphic connected-carrier activities through that branch with explicit losses of analytic radius and support weight. The number of Taylor coordinates is counted exactly, and the vacuum term is separated. This supplies a bounded one-step transport for the generated quantum activities. It does not assert contraction or stability of the analytic domains under repeated integration.

Notebook 21 localized the classical action and controlled one integration of it in independent current backgrounds. A new coarse effective action requires the substitution of its stationary current background. The following two results prove that substitution and its activity norm estimate. Their domain and weight losses are retained as inputs to the remaining quantum induction.

1 Strict support of the physical stationary background

Fix odd L, dimension four, color and the compatible periodic source geometry with distinct block interfaces. Current spacing is one, old fine spacing is M^{-1} with $M = L^k$, and next-coarse spacing is L. Write $V = e^{ia}$ and $W = e^{iw}$. These a, w are link angles. In current physical units the next-coarse potential is w/L; in next-coarse unit coordinates it is w. All formulas below use w, not w/L. Constants may depend on fixed L, but not on M or volume.

Group current coordinates into L-block cells, indexed by next-coarse cells. The retained nonpivot coordinate z and next-coarse datum w have fixed real dimensions per block

$$D_z = d(L^d - 1)\dim \mathfrak{g}, \qquad D_w = d\dim \mathfrak{g}, \qquad D = D_z + D_w = dL^d \dim \mathfrak{g}. \tag{1}$$

No fluctuation coupling has yet been inserted in z. Fixed-dimensional norm equivalences are included in all radii. Scalar Cauchy estimates use an actual coordinate product polydisc after this conversion.

We use the previously established ordinary source average and pivot chart, the physical gauge-zero branch obtained by the two minimum comparisons, and the strict scalar classical-action expansion. The flat retained precision has the uniform bounds

$$A_{M,0} = K_L^T(q_M + T_L^T T_L)K_L, \qquad C_{M,0} = A_{M,0}^{-1}, \qquad \gamma_* I \le A_{M,0} \le B_* I, \tag{2}$$

and both matrices have uniformly exponentially summable block rows and columns. Strict scalar support is available with any prescribed finite cardinality reserve, at the cost of a smaller uniform product radius. The result proved here is one uniform classical-background transport, not a claim of closure of repeated quantum RG steps.

1.1 A local affine pivot chart

At the identity background let

$$F_1(a) = \frac{1}{i}\log M_L(e^{ia}) = A_1 a + C_1(a), \qquad C_1(0) = DC_1(0) = 0. \tag{3}$$

The actual source normalization is

$$A_1 = LQ_L^{\mathrm{flat}}, \qquad A_1 h_1 = I, \qquad A_1 K_L = 0, \qquad h_1(c) = L^{d-1} I \text{ at } p(c). \tag{4}$$

The pivot coefficient in Q_L^{flat} is L^{-d}, hence its coefficient in A_1 is L^{1-d}. Nonpivot restriction of K_L is the identity; h_1 has only pivot entries.

Prescribe the next-coarse datum w and the retained absolute logarithmic coordinates z by solving

$$a = K_L z + h_1 w - h_1 C_1(a). \tag{5}$$

Each ordinary constraint equation reads its own pivot and no other pivot. Thus this is a family of independent finite-dimensional Lie-algebra-block implicit equations, with neighboring nonpivot values as parameters. Their derivative in the pivot at zero is invertible. There are only finitely many local types. A common small contraction therefore gives a holomorphic solution $a = \mathfrak{a}(w, z)$ satisfying

$$\begin{aligned} &F_1(\mathfrak{a}(w,z)) = w, && \mathfrak{a}(w,z)|_{\mathrm{nonpivot}} = z, \\ &\|\mathfrak{a}(w,z)\|_\infty \le C_L(\|w\|_\infty + \|z\|_\infty), && \mathfrak{a}(w,z) = K_L z + h_1 w + O((\|w\| + \|z\|)^2). \end{aligned} \tag{6}$$

Iteration does not spread to another pivot equation. The dependency neighborhood and all local analytic constants are fixed at L and independent of M and volume. This is the unique small constraint solution with the specified nonpivot values, equivalently the identity-background ambient source straightening evaluated at $K_L z + h_1 w$.

1.2 Scalar support and the exact flat mixed term

Let G be the actual averaged-contour exponential gauge action, in the coupling-free normalization, and define

$$T_M(w,z) = S_M(\mathfrak{a}(w,z)) + G(e^{i\mathfrak{a}(w,z)}). \tag{7}$$

The strict scalar support theorem applies to this fixed-range analytic substitution and the local gauge action. After fixed L-block regrouping it yields

$$T_M(w,z) = \sum_{X \neq \varnothing} T_X(w|_X, z|_X), \tag{8}$$

$$\mathcal{K} := \sup_i \sum_{X \ni i} e^{\sigma\tau(X) + A_{\mathrm{c}}|X|} \|T_X\|_\rho < \infty. \tag{9}$$

Here τ is graph Steiner length on the L-block graph, allowing auxiliary vertices, $\sigma > 0$, and $A_{\mathrm{c}} > 0$ is any prescribed finite reserve. The coordinate-polydisc radius ρ and $\mathcal{K}$ are uniform in M and volume. Every coefficient starts at degree at least two: the action does so, the chart vanishes at zero, and the gauge action has no constant or linear part at the identity.

Put $P_M = q_M + T_L^T T_L$. The full scalar action is stationary at zero, so only the linear chart enters its quadratic term:

$$T_M^{(2)}(w,z) = \tfrac{1}{2}\langle K_L z + h_1 w, P_M(K_L z + h_1 w)\rangle. \tag{10}$$

Thus

$$D_z T_M(w,z) = A_{M,0} z + B_M w + N_M(w,z),$$
$$B_M = K_L^T P_M h_1 = K_L^T q_M h_1, \qquad N_M(0,0) = DN_M(0,0) = 0. \tag{11}$$

The last simplification follows from $T_L h_1 = 0$, because the root contours are block internal and contain no pivot. The rectangular B_M has uniform exponential block bounds. The pairings here are counting pairings of the stated angle coordinates; no additional L^4 transpose factor is introduced. Replacing w by the next-coarse physical potential means substituting $w = La_{\mathrm{next}}$ explicitly.

1.3 An analytic mixed-slot bound from scalar supports

The supports in (8) are not uniformly bounded. The finite-range vertex lemma therefore cannot be applied directly to their gradient. In particular, crude polarization followed by $|X|^n$ slot counting can produce a factorial loss. Coordinatewise Cauchy gives the needed stronger conclusion.

Write $u = (w,z)$ in a fixed scalar basis. On the coordinate polydisc there are $D|X|$ variables. If $M_X = \|T_X\|_\rho$, its scalar Taylor coefficients satisfy

$$|c_{X,\beta}| \le M_X \rho^{-|\beta|}. \tag{12}$$

For one retained derivative coordinate α at a cell $i \in X$, differentiating the product of geometric series gives, for $t < \rho$,

$$\sum_{\beta \ge e_\alpha} \beta_\alpha |c_{X,\beta}| t^{|\beta|-1} \le \frac{M_X}{\rho}(1 - t/\rho)^{-D|X|-1}. \tag{13}$$

Sum the at most D_z output components. A symmetric multilinear Taylor coefficient distributes each scalar monomial coefficient among its ordered slots with multinomial weights summing to one. Consequently the sum of its ordered cell-slot operator norms is bounded by the same absolute scalar coefficient sum, with fixed norm-conversion constants absorbed in this bound. Every output and input-slot root lies in X, so their rooted Steiner length is at most $\tau(X)$.

For $0 < \kappa \le \sigma$, the supremum of the all-order positive row series is therefore bounded by

$$\mathcal{G}(t) = \frac{D_z}{\rho} \sup_i \sum_{X \ni i} e^{\kappa \tau(X)} M_X (1 - t/\rho)^{-D|X|-1}. \tag{14}$$

Choose $t_0 < \rho/2$ such that

$$D \log \frac{1}{1 - t_0/\rho} \le \frac{A_c}{2}. \qquad \text{Then} \qquad \mathcal{G}(t_0) \le \frac{2D_z \mathcal{K}}{\rho} =: M_g. \tag{15}$$

The supremum over output cells cannot be interchanged with the order sum. Instead let v_n be the supremum of the order-n mixed-slot row norm. Equation (14) gives the separate bound $v_n \le M_g t_0^{-n}$. Removing degrees zero and one gives the positive majorant for N_M:

$$\Phi(t) = \sum_{n \ge 2} v_n t^n \le \frac{M_g (t/t_0)^2}{1 - t/t_0}, \qquad \Phi'(t) \le \frac{8 M_g t}{t_0^2} \quad (0 \le t \le t_0/2). \tag{16}$$

This deliberate geometric factor makes the order/root-supremum bookkeeping valid. It leaves an analytic quadratic majorant uniform in refinement and volume.

The original X is a fixed coefficient carrier at the flat baseline, not an additional variable support. After substituting families for w, z, their union is the actual variable support. Internal rooted connections through X have already been paid by (14). This preserves strict dependence on the original next-coarse variables.

1.4 The support-valued stationary fixed point

For a prescribed finite $a_c \geq 0$, use

$$\|u\|_{\kappa,a_c;r} = \sup_i \sum_X e^{\kappa\tau(\{i\}\cup X)+a_c|X|} \sup_{\|w|_X\|<r} \|u_{i;X}(w|_X)\|. \tag{17}$$

The source w has singleton norm $e^{a_c}r$. For joint insertion in N_M, use the maximum of the separate w- and z-family norms and color slots by scalar coordinate and field type. The positive scalar majorant above already sums all such colors and types, while each scalar-component family is bounded by its vector-family norm. This does not identify that maximum with $\sum_X \|(w_X, z_X)\|$, which could be larger.

Linear exponential block bounds act in (17) by their weighted row sums. The rooted Steiner union inequality and $e^{a_c|\cup X_r|} \leq \prod_r e^{a_c|X_r|}$ allow the proved support-substitution lemma to use (16). Stationarity becomes

$$z = -C_{M,0}B_M w - C_{M,0}N_M(w,z). \tag{18}$$

Let c_0 bound the weighted norm of $C_{M,0}$ and choose $\ell_0 \geq \max\{1, \|C_{M,0}B_M\|_\kappa\}$ uniformly. Put

$$r_* \leq \min\left\{\frac{t_0}{2}, \frac{t_0^2}{16c_0M_g}\right\}, \qquad r \leq \frac{r_*}{2\ell_0 e^{a_c}}. \tag{19}$$

Decrease these radii further so that reconstruction and evaluation stay within the common prior chart and uniqueness neighborhoods. On the family ball $\|z\| \leq 2\ell_0 e^{a_c}r$, the map (18) is a contraction with constant at most $1/2$ and preserves that ball. Hence it has a unique holomorphic supported solution with

$$\|z_*\|_{\kappa,a_c;r} \leq 2\ell_0 e^{a_c}r, \qquad \|z_* + C_{M,0}B_M w\|_{\kappa,a_c;r} \leq C_{a_c}r^2. \tag{20}$$

Evaluation gives a genuine stationary solution on each finite regulator. Picard iteration defines an absolutely convergent rooted expansion with original w-cell leaves.

The finite-range chart $\mathfrak{a}$ can now be substituted by the finite-range support lemma. Define

$$a_*(w) = \mathfrak{a}(w, z_*(w)), \qquad H_{\mathrm{cur},M} = h_1 - K_L C_{M,0}B_M. \tag{21}$$

Then

$$\|a_*\|_{\kappa,a_c;r} \leq C_{a_c}r, \qquad \|a_* - H_{\mathrm{cur},M}w\|_{\kappa,a_c;r} \leq C_{a_c}r^2. \tag{22}$$

This $H_{\mathrm{cur},M}$ is the current-angle extension, distinct from the earlier original-fine extension. It has a uniform exponential rectangular block bound and $A_1 H_{\mathrm{cur},M} = I$. Its source gauge condition will follow by identifying the physical branch below; no gauge-zero conclusion is inferred from stationarity alone. Local exponentiation also supplies strict families for $e^{ia_*} - I$ and its inverse minus the identity. For activity transport the bound $s(r) = C_{a_c}r$ on the potential itself is sufficient.

1.5 Identification of the physical gauge-zero branch

For real small w, the already proved two-scale minimum comparison and rooted averaged-contour gauge construction give a branch $V_{\mathrm{old},*}(e^{iw}) = e^{ia_{\mathrm{old},*}}$ with

$$\begin{aligned} &M_L(V_{\mathrm{old},*}) = e^{iw}, && P_{V_{\mathrm{old},*}}(y,x) = 1, \\ &\|a_{\mathrm{old},*}\|_\infty \leq C_L\|w\|_\infty, && S_M(a) \geq S_M(a_{\mathrm{old},*}) \end{aligned} \tag{23}$$

for every real feasible a in the common small comparison ball. The actual gauge action is nonnegative on real unitary fields and vanishes at $P = 1$. Hence

$$S_M(a) + G(e^{ia}) \geq S_M(a_{\mathrm{old},*}) + G(V_{\mathrm{old},*}) \tag{24}$$

on that feasible neighborhood. The affine chart parametrizes every sufficiently small feasible field by its retained absolute coordinates. Thus $z_{\mathrm{old},*} = a_{\mathrm{old},*}|_{\mathrm{nonpivot}}$ is a critical point of $T_M(w,\cdot)$.

The scalar support expansion and Cauchy with fixed margins give

$$\|D_z^2 T_M(w,z) - A_{M,0}\|_\kappa \le C_L(\|w\|_\infty + \|z\|_\infty). \tag{25}$$

For each coefficient, contract its third derivative with the bounded direction (w,z); summing the second retained matrix index costs only $|X|$, paid by a cardinality spare. Its input/output distance is at most $\tau(X)$. Symmetry gives the column bound. This calculation differentiates the full scalar action, so both chart curvature and actual gauge curvature remain included. It does not use the ordinary rooted average's physical adjoint.

Choose a common real retained supremum ball and a data ball making (25) at most $\gamma_*/2$. The retained Hessian is positive on this convex ball, so there is at most one critical point. Both z_* from (18) and $z_{\mathrm{old},*}$ lie inside after a uniform shrink. Therefore

$$e^{ia_*(w)} = V_{\mathrm{old},*}(e^{iw}), \qquad M_L(e^{ia_*(w)}) = e^{iw}, \qquad P_{e^{ia_*(w)}}(y,x) = 1. \tag{26}$$

This identifies with a known gauge-zero minimum; a generic critical equation was not used to assert the gauge condition. Holomorphy and the finite-dimensional identity theorem extend these identities from the real ball to the common complex product domain. Real minimum inequalities are not extended as inequalities. Linearizing the last identity gives $T_L H_{\mathrm{cur},M} = 0$.

1.6 Transport conclusion and its domain limitation

For any prescribed finite cardinality reserve and sufficiently small positive κ, (22) is now a strict-support expansion of the actual physical stationary background in its original next-coarse angle variables. The radius is uniform in old refinement and volume. Output roots are L-blocks of current fields and input labels are next-coarse cells on the same block graph. If an activity uses another current-cell convention, its projection to this graph and the finite fiber multiplicity must be included explicitly; fixed regrouping is not a contraction estimate. Coefficients can depend on the fixed baseline and regulator. Strict variable support does not by itself assert volume consistency of a chosen decomposition.

There is a concrete obstruction to inferring an L^{-1} contraction of the angle supremum domain. Take small coarse pure-gauge data $W(y,y') = g_y g_{y'}^{-1}$ and set the current-site gauge to g_y throughout each L-block. Its current field is the identity internally and equals W on the corresponding block-boundary links. It has $P_V = 1$, ordinary average W and zero classical action. By the same minimum comparison and uniqueness it is the physical stationary representative. Its boundary angle is of order $\|w\|$, not $\|w\|/L$. The physical conversion $a_{\mathrm{next}} = w/L$ does not change that fact about the angle-domain map.

This theorem supplies one uniform step and an explicit small-domain bound $s(r) = C_a r$ for substitution in supplied analytic activities. Preservation of a quantum activity norm, any associated radius loss, and a compatible common domain under repeated integrations require separate proofs. No running-coupling, mixed-history, large-field or interacting continuum conclusion follows here.

2 Strict-support transport of analytic quantum activities

We now substitute the actual stationary current background into a supplied family of quantum activities. This is a bounded finite-step map. Its input has only one fixed positive support-cardinality reserve; we do not assume an arbitrary reserve for quantum activities. The number of Taylor coordinates is proportional to the original carrier size, and we explicitly pay that cost. Constants are uniform in the old refinement M and compatible periodic volume, with fixed L.

2.1 Carrier geometry and the precise analytic inputs

Let Λ be the input cell graph and Λ' the next field's cell graph. A projection $\pi : \Lambda \to \Lambda'$ has fibers of size at most F and sends every nonempty connected input carrier to a connected set. For ordinary L-cube blocking, $F = L^d$. Identity projection is allowed. Write d' for graph distance on Λ' and define

$$S_\delta = \sup_{z\in\Lambda'} \sum_{j\in\Lambda'} e^{-\delta d'(z,j)} \le \sum_{j\in\mathbb{Z}^d} e^{-\delta|j|_1} < \infty, \qquad \delta > 0. \tag{27}$$

The bound is uniform on compatible tori: choose one minimal-distance integer representative of every periodic site. Short periodic directions do not enlarge the sum.

Each input cell carries at most D scalar complex background coordinates. For a nonempty connected $H \subset \Lambda$, suppose $E_H(a|_H)$ is bounded and holomorphic on

$$|a_i^\alpha| < b, \qquad i \in H, \quad 1 \le \alpha \le D, \tag{28}$$

and put $e_H = \|E_H\|_b$. The only activity hypothesis is

$$\sup_{i\in\Lambda} \sum_{H\ni i} e^{a_0|H|} e_H \le \epsilon, \qquad a_0 > 0 \text{ fixed}. \tag{29}$$

Here D counts background scalar coordinates, not automatically retained Gaussian coordinates. A product of Euclidean fiber balls of radius b_E contains the scalar polydisc of radius $b_E/\sqrt{D}$; make this fixed norm conversion before applying the formulas below.

The actual stationary potential on the next-field product domain of radius r has the strict support expansion

$$a_i^\alpha(w) = \sum_{\varnothing\ne X\subset\Lambda'} u_{i,\alpha;X}(w|_X), \qquad u_{i,\alpha;X}(0) = 0, \tag{30}$$

$$\sup_{i,\alpha} \sum_X e^{\kappa\tau(\{\pi(i)\}\cup X)} \|u_{i,\alpha;X}\|_r \le s(r) \le C_a r, \qquad \kappa > 0. \tag{31}$$

Here τ is graph Steiner length, allowing auxiliary vertices. The preceding stationary-background theorem supplies this estimate; its stronger support-cardinality weight is not needed here. A maximum-coordinate bound on its full current-field block fiber implies (31) for each scalar component, without summing over the fiber.

If only the total $a_i(0)$ vanishes, subtract each summand's constant. Absolute convergence preserves the total and increases s by at most a factor two. Incorporate that factor into C_a. For a nonzero reference potential, center both (28) and (30) at that reference and use the available distance to the analytic boundary in place of b.

Remark 2.1 (The present physical indexing). The normalized Gaussian activities supplied by Notebooks 18 and 21 are already indexed by L-blocks of the current unit lattice. The next-coarse w labels are the same block index graph. The present physical substitution therefore uses

$$\pi = \mathrm{id}, \qquad F = 1, \qquad D = D_{\mathrm{bg}} = dL^d \dim \mathfrak{g}.$$

The last dimension includes all current background links assigned to the block; it differs from the retained Gaussian dimension $d(L^d - 1)\dim\mathfrak{g}$. After substitution, these labels are unit cells of the next current lattice. A further projection for the following integration is a distinct operation, discussed below. It must not be counted as an additional gain during this substitution.

2.2 Bounded transport with an explicit loss budget

Choose $a_{\mathrm{out}}, \mu \geq 0$ and $\delta > 0$, and set

$$\beta = a_{\mathrm{out}} + \mu, \qquad t = \frac{s(r)}{b} < 1, \qquad \alpha_D(t) = D \log \frac{1}{1-t}. \tag{32}$$

For a connected output carrier K, let $\ell(K)$ denote metric minimum spanning-tree length. Define

$$\|E'\|_{a_{\mathrm{out}},\mu;r} = \sup_{z \in \Lambda'} \sum_{K \ni z} e^{a_{\mathrm{out}}|K| + \mu \ell(K)} \|E'_K\|_r \,. \tag{33}$$

Theorem 2.2 (Finite-step activity transport)**.** *Assume* (29)–(31) *and*

$$\beta + \delta \leq \kappa, \qquad \beta + \alpha_D(t) \leq a_0. \tag{34}$$

There are strictly supported holomorphic activities $E'_K(w|_K)$ on nonempty connected $K \subset \Lambda'$, with $E'_K(0) = 0$, such that

$$\sum_H E_H(a(w)|_H) = \mathrm{Vac}_\Lambda + \sum_K E'_K(w|_K), \qquad \mathrm{Vac}_\Lambda = \sum_H E_H(0), \tag{35}$$

$$\|E'\|_{a_{\mathrm{out}},\mu;r} \leq F S_\delta \epsilon. \tag{36}$$

All regroupings converge absolutely. The bound is independent of M and finite periodic volume. It is not a contraction assertion.

A concrete choice with strictly positive output weights is

$$\beta = \min\{a_0/4, \kappa/4\}, \qquad a_{\mathrm{out}} = \mu = \beta/2, \qquad \delta = \kappa/4,$$
$$\frac{s(r)}{b} \leq \min\{1/2, a_0/(8D)\}. \tag{37}$$

Indeed $\alpha_D(t) \leq 2Dt \leq a_0/4$. Thus a sufficient radius is

$$r \leq \frac{b}{C_a} \min\{1/2, a_0/(8D)\}, \tag{38}$$

together with the stationary-background theorem's own radius restriction. All centering and norm-conversion constants are included in b or C_a. Nothing requires $C_a < 1$.

Proof. Expand each activity in its actual $D|H|$ scalar coordinates:

$$E_H(a|_H) = \sum_{\nu \in \mathbb{N}^{D|H|}} c_{H,\nu} a^\nu, \qquad |c_{H,\nu}| \leq e_H b^{-|\nu|}. \tag{39}$$

Coordinatewise Cauchy on smaller polydiscs, followed by a limit, gives the coefficient bound. There is no dimension-free absolute Taylor norm being assumed.

For every scalar occurrence in a monomial a^ν, insert (30). Its chosen support X inherits the root $\pi(i)$ of that scalar coordinate. For each such rooted set, fix a deterministic minimizing connected tree $T(i, X)$ containing $\{\pi(i)\} \cup X$. Ties can be resolved by any fixed finite-volume rule. For a decorated term define

$$K = \pi(H) \cup \bigcup_{\text{occurrences}} \mathrm{Vert}\, T(i, X), \qquad T_{\mathrm{tot}} = \sum_{\text{occurrences}} \tau(\{\pi(i)\} \cup X). \tag{40}$$

The original projected carrier is retained even if some of its variables are absent from the monomial. It is connected, and every attached tree has a root in it. Consequently

$$|K| \le |\pi(H)| + T_{\mathrm{tot}}, \qquad z \in K \implies d'(z, \pi(H)) \le T_{\mathrm{tot}}. \tag{41}$$

A tree with q edges adds at most q vertices after attachment at its root. A vertex on an attached tree is at distance at most that tree's length from $\pi(H)$. Overlaps and periodic wrapping can only improve these inequalities.

The decorated function reads only $w|_K$. Its true variable set may be smaller: K is a declared connected carrier, not a claim of minimal dependence. A nonconstant monomial is a product of factors vanishing at zero and hence also vanishes at zero. Define E'_K as the sum of these nonconstant terms having carrier K. Set aside the constant Taylor terms as Vac_Λ.

For every connected lattice vertex set K,

$$\ell(K) = |K| - 1, \qquad e^{a_{\mathrm{out}}|K| + \mu\ell(K)} \le e^{\beta|K|}. \tag{42}$$

Indeed the induced connected graph has a unit-edge spanning tree, while every metric tree has $|K| - 1$ edges of distance at least one. Graph Steiner length is used for the attached rooted sets because deleting intermediate roots from a terminal metric spanning tree need not decrease its length. At the output all connecting vertices are retained, so (42) applies exactly.

Put $h = |\pi(H)|$ and $k = |H|$. For a fixed output anchor z, (41) gives

$$\mathbf{1}_{\{z \in K\}} e^{a_{\mathrm{out}}|K| + \mu\ell(K)} \le e^{\beta h} e^{-\delta d'(z, \pi(H))} \prod_{\text{occurrences}} e^{(\beta+\delta)\tau(\{\pi(i)\} \cup X)}. \tag{43}$$

If $z \notin K$ this is automatic; otherwise insert $1 \le e^{\delta(T_{\mathrm{tot}} - d'(z, \pi(H)))}$. The first condition in (34) lets us sum each scalar occurrence's support using (31), at cost $s(r)$. The exact sum over scalar multiindices is

$$\sum_{\nu \in \mathbb{N}^{Dk}} t^{|\nu|} = (1-t)^{-Dk} = e^{\alpha_D(t)k}. \tag{44}$$

Omitting $\nu = 0$ replaces the last expression by $e^{\alpha_D(t)k} - 1$. Thus the full contribution from a fixed H is bounded by

$$e_H e^{\beta h + \alpha_D(t)k} e^{-\delta d'(z, \pi(H))}. \tag{45}$$

Since $h \le k$, the second condition in (34) pays the two exponential costs from the given reserve $a_0 k$. Finally,

$$e^{-\delta d'(z, \pi(H))} \le \sum_{i \in H} e^{-\delta d'(z, \pi(i))}. \tag{46}$$

Summing (45) over H therefore gives

$$\sum_i e^{-\delta d'(z, \pi(i))} \sum_{H \ni i} e^{a_0|H|} e_H \le F S_\delta \epsilon, \tag{47}$$

which proves (36).

This positive majorant also justifies every substitution and regrouping. At finite volume, summing over output anchors bounds the total absolute series. Normal convergence on smaller product polydiscs proves holomorphy. No order sum has been interchanged with an output supremum: for each fixed H the scalar series is summed exactly in (44), before taking the anchored carrier sum. In particular, no crude $|H|^n$ slot count with a missing factorial is hidden in the argument. □

2.3 Vacuum and the smaller-radius improvement

The vacuum has the explicit density assignment

$$v_i = \sum_{H \ni i} \frac{E_H(0)}{|H|}, \qquad \mathrm{Vac}_\Lambda = \sum_i v_i, \qquad |v_i| \le \epsilon, \qquad |\mathrm{Vac}_\Lambda| \le |\Lambda|\epsilon. \tag{48}$$

Grouping these densities into a block costs at most F.

Corollary 2.3 (Vacuum-subtracted radius bound)**.** *If the hypotheses of Theorem 2.2 hold with*

$$\rho = a_0 - \beta - \alpha_D(t) > 0, \tag{49}$$

then also

$$\|E'\|_{a_{\mathrm{out}},\mu;r} \le F S_\delta \epsilon\, \alpha_D(t) C_\rho, \qquad C_\rho = \sup_{k \ge 1} k e^{-\rho k} \le \frac{1}{e\rho}. \tag{50}$$

Proof. For nonconstant terms use $e^{\alpha_D(t)k} - 1 \le \alpha_D(t) k e^{\alpha_D(t)k}$ in place of (44). The spare $e^{-\rho k}$ pays the factor k. The remaining rerooting sum is unchanged. □

One may take the smaller of (36) and (50). This is improved smallness on a strictly smaller field domain. It does not prove a contraction at a common radius under repeated RG steps.

2.4 An optional connected-carrier compression estimate

Lemma 2.4 (Connected carriers under cube projection)**.** *For odd $L \ge 5$, put*

$$r_L = (L-1)/2, \qquad s_L = \lfloor (r_L - 1)/2 \rfloor, \qquad k_L = s_L + 1. \tag{51}$$

If H is connected and π_L is L-cube projection, then

$$|\pi_L(H)| \le 2^d \max\{1, |H|/k_L\} \le 2^d + (2^d/k_L)|H|. \tag{52}$$

The estimate holds on compatible rectangular tori.

Proof. Choose a maximal subset of H separated by at least r_L in the periodic supremum metric. The supremum balls of radius $r_L - 1$ around its J centers cover H. Each such ball has side length $2r_L - 1 < L$ and meets at most 2^d coarse blocks, so $|\pi_L(H)| \le 2^d J$. If $J = 1$ the result follows. If $J \ge 2$, the graph-distance balls of radius s_L about the centers are disjoint, since $2s_L < r_L$. For each center, a path in H to another center contains at least $s_L + 1$ distinct vertices before leaving its ball. Hence $Jk_L \le |H|$. Identifications in short periodic directions only reduce the block counts; the same separation and path argument uses periodic distances. □

Corollary 2.5 (Transport with a proved carrier gain)**.** *Suppose the projection in Theorem 2.2 satisfies $|\pi(H)| \le c_0 + c_1|H|$ on connected carriers. The second condition in (34) may be replaced by*

$$c_1\beta + \alpha_D(t) \le a_0, \tag{53}$$

and the resulting bound is

$$\|E'\|_{a_{\mathrm{out}},\mu;r} \le e^{\beta c_0} F S_\delta \epsilon. \tag{54}$$

The smaller-radius estimate also holds with the factor $e^{\beta c_0}$ and $\rho = a_0 - c_1\beta - \alpha_D(t) > 0$.

Proof. Keep $h = |\pi(H)|$ in (45), and use $\beta h \le \beta c_0 + c_1\beta|H|$. All other parts of the proof are unchanged. □

Lemma 2.4 provides $c_0 = 2^d$, $c_1 = 2^d/k_L$. A sufficiently large fixed L can therefore help pay stronger cardinality weights. But the attached background trees must still obey $\beta + \delta \leq \kappa$ in the graph used by the substitution. A larger spatial exponent cannot be assumed without a separate regrouping proof and its accompanying cardinality and domain costs. The constant $e^{\beta 2^d} L^d S_\delta$ can be large.

In the actual convention of Remark 2.1, current substitution uses identity projection. Afterwards, its connected carriers are unit-cell carriers of the next current field w. Regrouping them into the following integration's L-block Gaussian cells is a legitimate new projection. It must include that later pivot-chart enlargement and background-domain budget; it is not a second projection hidden in (35).

2.5 What the estimate does and does not establish

Connectedness of the input carrier is necessary for this cardinality-only hypothesis. Without it, an activity reading two sites at arbitrarily large separation can have fixed cardinality norm, while any connected output carrier containing both has size at least their distance plus one. A spatial input norm could pay this cost, but it is not part of (29).

There is no automatic contraction: a constant singleton activity $E_{\{i\}} = c$ at each old cell gives $L^d c$ after summation in a full coarse block. Even after vacuum subtraction, an identity background map and a linear singleton activity have no strict contraction at the same analytic radius. The gain in (50) is explicitly a radius loss.

For example, a Gaussian output reserve $a_0 \leq 1/2$ cannot simply be re-used as an interaction input demanding exponent 4. One must verify a compression/enlargement budget, a suitable stationary-background exponent, and the next domain. The stationary branch has a one-step radius with an L-dependent constant; no supremum-domain contraction by L^{-1} has been proved. These facts leave a genuine induction interface.

Every transported coefficient is strictly a function of its declared w variables. Its fixed baseline may still depend on M and the finite regulator, and a deterministic choice of connecting trees need not give coefficientwise volume consistency. The uniform bounds alone imply no thermodynamic consistency.

Finally, the theorem transports supplied quantum activities through the matched classical stationary background. It does not identify the full interacting quantum stationary point with that classical branch, absorb inherited interactions into a new minimization, or control running couplings, large-field histories, or observable insertions. It supplies an exact finite-step bridge whose coefficient, support, volume, and analytic-radius costs are all displayed.

3 The remaining quantum recurrence

The actual stationary substitution is now represented by strictly supported functions of the next-coarse variables. The connected activities supplied by Notebook 21, and its separate normalization terms after a common reduction of their support exponent, satisfy the input form of the transport theorem. Thus their sum can be expressed as local next-coarse activities on the explicitly smaller analytic ball. This is a completed finite-step interface.

For the specific convention used in Notebook 21, the integration cells are already the next-coarse block cells. That substitution uses the identity projection. Projection into another set of L-blocks occurs only when preparing these activities for the following integration. It cannot be counted twice.

A repeated quantum construction requires more. The generated activity norm has a fixed small support exponent, whereas the Gaussian integration theorem asks for a stronger weighted input. The connected-carrier estimate explains a possible geometric gain at the later projection, but its constants and the new fluctuation-chart enlargement must be included. Notebook 24 makes that inherited-interaction budget explicit; Notebook 26 meets it using a fixed auxiliary grouping and recovers a common positive carrier exponent. The smaller analytic radius in the transport theorem must also

be replaced by a proved stable domain recursion. The explicit pure-gauge example above rules out obtaining that recursion merely by declaring that every link angle contracts by L^{-1} in this rooted representative. Notebook 25 constructs a smaller pre-root-gauge representative and identifies the analytic composition germ for an invariant total functional, with its remaining activity and integral domain requirements kept explicit.

Finally, the vacuum, marginal gauge-action coefficient and irrelevant interactions need their appropriate extractions and scaling estimates. A bounded norm map is insufficient: vacuum singleton terms already grow by block volume. Large-field complements, source insertions and the continuum reconstruction requirements of Notebook 01 remain part of the full existence argument.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 15–21 in this workspace. The physical stationary branch, exact current pivot chart, retained inverse, scalar support construction and connected integration activities are the principal inputs.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. **109** (1987), 249–301, original article copy.

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Commun. Math. Phys. **116** (1988), 1–22, original article copy.

[4] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.